Intelligent Whisper-Based Call Recording Analysis System for CSR Emotion Detection and Automated Intervention Recommendations Using MFCC and Lightweight Machine Learning

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A Thesis Presented to
The Faculty of the College of Computer Studies
Tarlac State University
Tarlac City

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In Partial Fulfillment
of the Requirements for the Degree
Computer Science

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by:

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TARLAC STATE UNIVERSITY
COLLEGE OF COMPUTER STUDIES

TCP FORM 14

APPROVAL SHEET

This Thesis of **Allan T. Galivo, Flory Mae Gader,** and **Romina Jane R. Salazar** entitled "**Intelligent Whisper-Based Call Recording Analysis System for CSR Emotion Detection and Automated Intervention Recommendations Using MFCC and Lightweight Machine Learning**", which is prepared and submitted in partial fulfillment of the requirements of the degree **Bachelor of Science in Computer Science** is hereby approved and accepted.

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ABSTRACT

Insert your abstract here. This portion is not to be indented and should be clear, concise, and complete. An abstract tells the reader the reason for writing the thesis or capstone project, the current approaches and gaps in the literature, the different objectives or aims of the research, the methodology used in attaining the objectives, the main findings, and end it with the main conclusion including the implications of the research. Limit the abstract writing to 350 words.

DEDICATION

This portion is optional but perhaps you have someone or some people who have inspired you to push on with your studies? A dedication would be a fitting way to acknowledge their impact on your success.

ACKNOWLEDGMENT

The road to this point in your studies couldn't have been travelled alone. Along the way, someone somewhere helped you. This is your chance to thank them.

By the way, exercise the liberty to be personal to reflect the sincerity of your gratitude…

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Chapter 1

INTRODUCTION

A. Project Context

Customer Service Representatives (CSRs) frequently encounter emotionally demanding interactions that can lead to stress, fatigue, and performance decline. Occupational psychology research identifies affective load, vocal stress, and emotional fatigue as common risks in communication-heavy environments such as call centers [9], [10]. However, existing BPO Quality Assurance (QA) tools focus primarily on customer sentiment and script adherence, lacking the ability to analyze CSR emotional state based on voice characteristics.

Speech Emotion Recognition (SER) literature shows that emotional cues typically categorized into the six basic emotions (happiness, sadness, anger, fear, disgust, surprise) can be detected from acoustic patterns in speech using MFCC and machine learning [3], [5], [11], [12]. These cues align with psychology's established markers of affective arousal, vocal tension, and prosodic regulation.

Whisper, a transformer-based Automatic Speech Recognition (ASR) model, provides high transcription accuracy even in noisy call center environments, supporting downstream emotion analysis [14]. Lightweight machine learning algorithms such as Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN) provide reliable classification performance for SER tasks [11], [12].

Despite these advancements, no existing BPO-centered system integrates Whisper ASR, MFCC-based emotion extraction, and lightweight ML to assess CSR emotional states and provide psychology-aligned, non-diagnostic recommendations for supervisors. This study addresses that gap.

B. Purpose and Description

The purpose of this study is to develop an intelligent system that analyzes CSR call recordings using Whisper Automatic Speech Recognition (ASR), Mel-Frequency Cepstral Coefficients (MFCC), and lightweight machine learning classifiers to identify emotional cues and provide psychology-aligned supervisory recommendations. Research in Speech Emotion Recognition (SER) has shown that the six basic emotions happiness, sadness, anger, fear, disgust, and surprise can be reliably estimated from acoustic features such as pitch, tone, and intensity, which MFCC effectively captures [1], [3], [11]. These features correlate with psychological constructs including affective arousal, vocal tension, and prosodic regulation, commonly used in occupational voice assessment [5], [9]. However, existing BPO quality tools focus primarily on customer sentiment rather than the emotional state of CSRs, leaving a significant gap in supportive systems for agent well-being.

The proposed system addresses this gap by integrating Whisper ASR for accurate transcription of CSR audio in noisy call center environments [14], and MFCC for emotional feature extraction. These features are then classified using Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN), which have demonstrated strong performance in SER tasks [11], [12]. The system translates emotional

indicators into three occupational emotional states: Regulated Affective State, Elevated Affective Load, and High Affective Load, based on established markers of vocal stress and affective activation. These classifications are used to generate non-diagnostic, psychology-aligned recommendations for supervisors, such as maintaining workload, providing supportive coaching, or offering short rest to prevent emotional fatigue.

A secure, offline dashboard will allow supervisors to upload recordings, review transcriptions, view emotional state results, and receive recommended interventions. By combining accurate speech recognition, validated emotional acoustic features, and interpretable machine learning methods, the study aims to create a practical and ethical tool that enhances CSR well-being, supports data-driven decision-making, and improves performance management in the BPO industry.

C. Objectives

Main Objective

The main objective of this study is to develop an affordable, AI-powered Whisper-Based Call Recording Analysis System that analyzes CSR speech, detects emotional cues using MFCC and lightweight machine learning, and provides psychology-aligned supervisory recommendations to support performance management and emotional wellness in Business Process Outsourcing (BPO) environments

Specific Objectives

- To implement and optimize a speech-to-text module using Whisper ASR to accurately transcribe CSR audio extracted from call recordings, providing reliable textual and acoustic inputs for emotion analysis.
- To implement an acoustic feature extraction method using Mel-Frequency Cepstral Coefficients (MFCC) to capture psychoacoustic properties of CSR speech that correlate with emotional states such as tension, calmness, frustration, and fatigue.
- To select, train, and evaluate lightweight machine learning algorithms Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN)—to classify emotional cues detected through MFCC features.
- To classify CSR vocal emotional indicators into three psychology-aligned emotional states, namely:
    A. Regulated Affective State (stable emotional regulation)
    B. Elevated Affective Load (mild to moderate vocal tension or stress)
    C. High Affective Load (heightened stress markers or emotional fatigue)
- To generate non-diagnostic supervisory recommendations based on the classified emotional state, such as maintaining workload, providing supportive coaching, or recommending a rest period or decompression interval.

- To design and develop the operational system capable of the following core functions:

  A. Emotion and Workload Regulation Analysis

     - Analyzing CSR speech for emotional markers such as vocal tension, prosodic dysregulation, affective arousal, or fatigue.
     - Presenting emotional state classifications using validated psychological terminology.
     - ○ Providing supervisors with recommended actions aligned with occupational well-being.

  B. Supervisor / Admin Dashboard Functions

     - Allowing supervisors and administrators to upload CSR call recordings for analysis.
     - Displaying Whisper-generated transcripts, emotional markers, and emotional state classifications.
     - Showing historical results, trends, and emotional performance over time.
     - Enabling administrators to manage user accounts, audit system usage, and configure settings.

  C. CSR / User Functions (If permitted by company policy)

    - Allowing CSRs to view their personal emotional analysis results.
    - Providing access to feedback or system-generated recommendations such as coaching suggestions or rest reminders.
- To ensure that the system operates fully offline to maintain data privacy, prevent information leakage, and support secure deployment within BPO facilities.
- To evaluate system performance using accuracy, precision, recall, and F1-score, determining its reliability as an objective, data-driven CSR emotional monitoring tool for supervisory decision-making.

D. Scope and Limitation

The scope of this study is limited to the development of an offline, CSR-centered emotional analysis system that processes post-call audio recordings and focuses exclusively on the Customer Service Representative's (CSR) vocal signals extracted from telephony-generated call files. Using Whisper for automatic speech recognition, the system transcribes CSR speech with robustness to noise, accents, and multilingual variations commonly encountered in BPO environments. Emotional features are extracted using Mel-Frequency Cepstral Coefficients (MFCC) to capture psychoacoustic indicators associated with vocal tension, affective arousal, calmness, frustration, and fatigue.

These features serve as input to lightweight machine learning classifiers Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN) which classify CSR emotional cues into three psychology-aligned occupational states: Regulated Affective State, Elevated Affective Load, and High Affective Load. Each classification is paired with non-diagnostic supervisory recommendations such as maintaining workload, providing supportive coaching, or recommending short rest. The system includes an offline dashboard for supervisors that presents transcriptions, emotional classifications, and historical trends, while also supporting administrative functions such as account management and usage auditing.

This study intentionally excludes the development of a full Quality Assurance suite; it does not perform script adherence scoring, compliance checks, silence detection, customer sentiment analysis, interaction quality metrics, or telephony management functions. The system analyzes only CSR audio and supports only post-call evaluation, not real-time monitoring.

The study is subject to several limitations. It is not a psychological, clinical, or medical diagnostic tool and relies exclusively on acoustic emotional markers, excluding facial expressions, physiological signals, and deeper linguistic semantic analysis. System accuracy depends heavily on audio quality, the clarity of CSR speech, background noise, microphone characteristics, and the availability and diversity of labeled emotional speech datasets.

Hardware constraints may also affect Whisper transcription speed, especially on low-specification devices, making real-time deployment impractical. Furthermore, accent variability, code-switching, and hybrid dialects may influence transcription and emotion classification accuracy. Finally, emotional state classifications and generated supervisory recommendations are advisory in nature; supervisors retain full authority over final decisions, and the outputs must be interpreted within these methodological and technical limitations.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

A. Discussion of Models

The core computational problem addressed in this study is the absence of an integrated system capable of analyzing CSR call recordings to detect emotional cues and classify emotional states in a way that supports effective supervisory decision-making. Existing speech analytics tools typically focus on customer sentiment, transcription quality, or general acoustic metrics, but very few provide a unified, CSR-centered pipeline for identifying emotional strain and vocal stress. This gap in current systems results in undetected emotional fatigue among call center agents and highlights the need for a specialized, automated approach that prioritizes the well-being of the CSR.

To address this challenge, the proposed system integrates three main components:

- Whisper for Automatic Speech Recognition (ASR)

- Mel-Frequency Cepstral Coefficients (MFCC) for acoustic feature extraction o
- Lightweight machine learning classifiers (SVM, Random Forest, KNN) for emotion classification.

Whisper is utilized due to its robustness in noisy and multilingual environments typical of call center operations, enabling accurate transcription that preserves important prosodic information such as pitch, tone, and rhythm. This transcription is then processed using MFCC extraction to capture psychoacoustic properties of speech that correspond to emotional characteristics such as tension, agitation, calmness, and fatigue. The study evaluates three machine learning models commonly used in Speech Emotion Recognition. SVM is known for its strong boundary-based performance when separating emotional states using MFCC features. Random Forest is selected for its resilience to noisy data and its ability to model nonlinear patterns, which is beneficial for real-world call center audio. KNN is included as a simple yet interpretable baseline model, offering competitive performance for smaller datasets despite scalability limitations. These classifiers are assessed based on their ability to differentiate emotional cues and map them to the system's defined emotional categories.

A key contribution of this model architecture is the translation of MFCC-derived emotional indicators into three psychology-aligned occupational emotional states: Regulated Affective State, Elevated Affective Load, and High Affective Load. These categories reflect levels of emotional regulation, vocal stress, and affective activation

without implying clinical diagnosis, allowing supervisors to interpret results meaningfully. Although similar ASR-SER pipelines exist, none are specifically designed for CSR analysis or incorporate psychology-informed interpretations intended for workplace intervention. By combining Whisper ASR, established acoustic feature extraction, and lightweight machine learning, the proposed system offers a computationally efficient and context-specific model that addresses the unique emotional monitoring needs of BPO environments.

Mel Frequency Cepstral Coefficients Algorithm

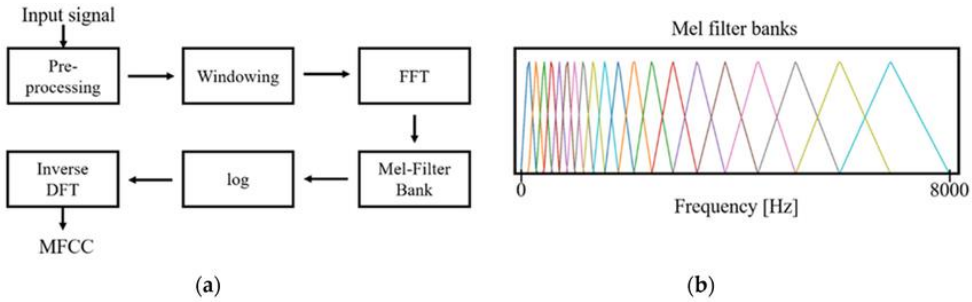

Figure 1. Flowchart of the Mel Frequency Cepstral Coefficients Algorithm

Mel-Frequency Cepstral Coefficients (MFCC) are features taken from a person's voice that help identify patterns related to emotion, such as stress, calmness, anger, or fatigue. The idea behind MFCC is simple: instead of analyzing the entire sound directly, the system breaks the voice into very small pieces and looks at how the energy of different frequencies changes over time. This is important because human hearing does not perceive all frequencies equally—we hear low and mid frequencies more clearly than high ones. MFCC uses this principle to extract only the most meaningful sound information from speech.

The process begins by preparing the audio and cutting it into small frames. Each frame is converted from raw sound waves into frequency information using the Fast Fourier Transform (FFT). Then, the frequencies are passed through Mel filter banks, which mimic how the human ear responds to sound. These filtered values are turned into log energies and then transformed using the Discrete Cosine Transform (DCT)

to produce the final MFCCs. These values represent the "shape" of the voice signal and reflect emotional characteristics in the speaker's tone.

The essential calculation for producing MFCCs is shown below. It is the step where the log Mel-filter bank energies are converted into cepstral coefficients:

$$C(n) = \sum_{k=1}^{K} \log(S(k)) \cos\left[n\left(k - \frac{1}{2}\right)\frac{\pi}{K}\right]$$

Where:

- $S(k)$is the log energy from the Mel filterbank
- $K$is the number of filters
- $C(n)$is the MFCC value produced

This formula is the heart of the MFCC method. It turns complex frequency information into simple numerical features that machine learning models—such as SVM, Random Forest, and KNN—can use to identify emotional signals in a CSR's voice.

Random Forest Algorithm

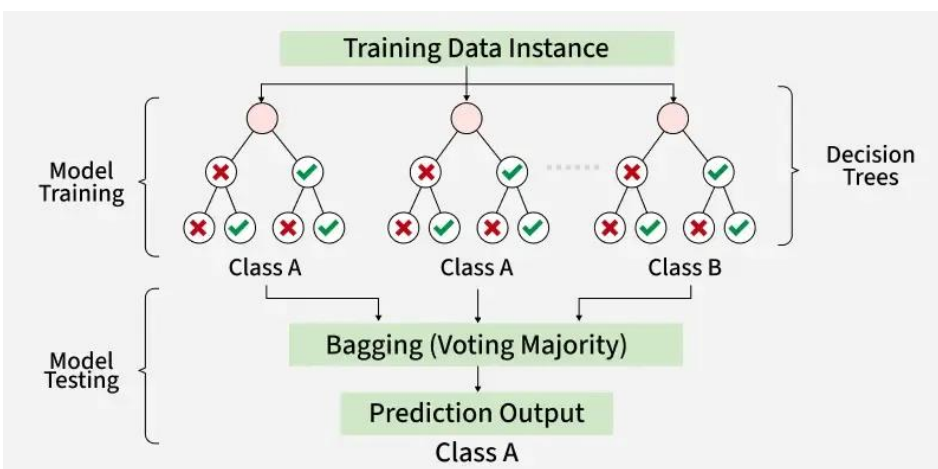

Figure 2. Model of Random Forest

Random Forest is an algorithm that uses many decision trees to classify emotions. Each tree makes a prediction, and the Random Forest chooses the most common answer. This makes RF very stable, especially for noisy audio like call center recordings, because even if one tree makes a mistake, the others correct it.

Key Random Forest Formula (Majority Voting)

$$\hat{y} = \text{mode}(h_1(x), h_2(x), \dots, h_n(x))$$

Where:

- $h_i(x)$= prediction of the i-th tree
- $\hat{y}$= final predicted emotion

This formula shows how the forest chooses the most frequent result across all trees.

Support Vector Machine Algorithm

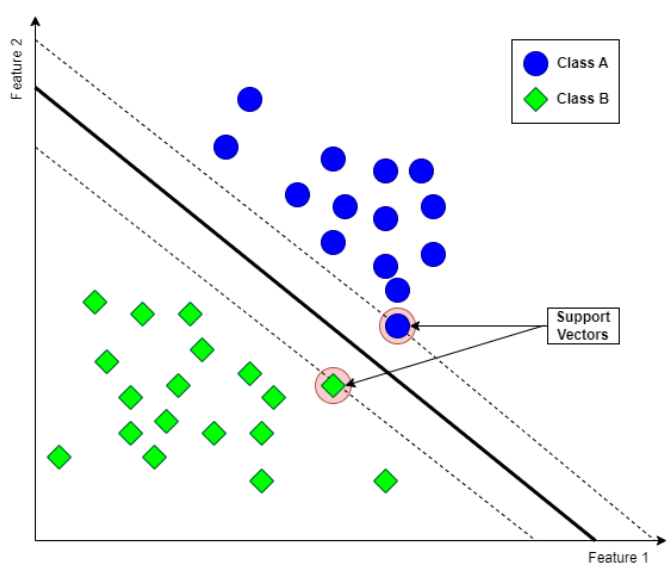

Figure 3. Model of Support Vector Machine

Support Vector Machine is a machine learning method that separates different emotion classes such as calm, stressed, or agitated by finding the best boundary between them. SVM looks for the line or plane that creates the largest margin between emotional categories so that the model can distinguish them more reliably. This makes SVM effective for MFCC features, which often show clear separations when emotions differ.

✓ Key SVM Formula (Decision Boundary)

$$w^T x + b = 0$$

Where:

- w= weight vector
- x= input features (MFCCs)
- b= bias

This formula represents the boundary SVM learns to separate emotions.

K-Nearest Neighbors (KNN) Algorithm

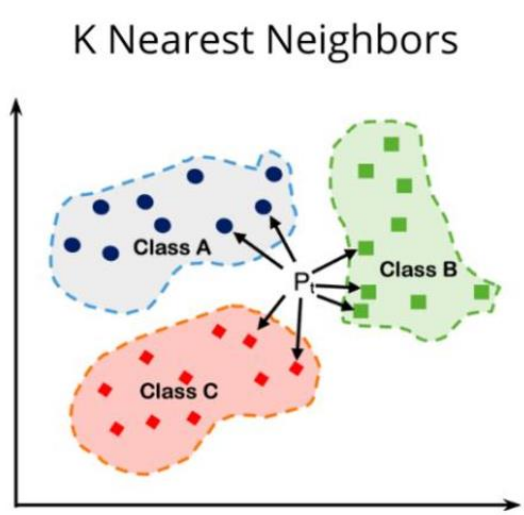

Figure 4. Model of K-Nearest Neighbors (KNN)

KNN is the simplest among the three. It classifies a new voice sample by looking at the closest existing samples in the dataset. The emotion label is determined by whichever group the majority of its nearest neighbors belong to. This method works well when emotional differences in MFCC features are clear and clustered.

$$d(x, x_i) = \sqrt{\sum (x - x_i)^2}$$

Where:

- x= new sample
- x_i= one of the known samples
- d= distance

KNN chooses the emotion class of the closest neighbors based on this distance.

Table 1. Functionality and Feature Matrix

Software	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Intelligent Whisper-Based CSR Call Recording Analysis System for Emotion Detection and Intervention Recommendation Using MFCC and Lightweight Machine Learning	✓	✓	✓	✓	✓	✓	✓	✓
SenseVoice (FunAudioLLM)	✓	✓	✓	✓	✓	✓	✓	✕
OpenAI Whisper	✓	✓	✕	✕	✓	✕	✓	✕

VOSK	✓	✓	✕	✕	✓	✓	✓	✓
SpeechBrain	✓	✓	✓	✓	✓	✕	✓	✕
openSMILE	✕	✕	✓	✓	✓	✕	✓	✕
Streamlit Real-Time SER Bot	✓	✓	✕	✓	✓	✓	✓	✕

Feature Definitions:

1. Speech-to-Text (Automatic Speech Recognition)
The system converts spoken CSR audio into text using Whisper, a transformer-based ASR model known for its high transcription accuracy and reliability in complex acoustic conditions. Accurate transcription improves the quality of downstream emotion analysis, as supported by SER surveys and ASR–SER integration studies.

2. Multilingual Support
Whisper provides robust transcription across multiple languages and accents, an important capability for call centers that frequently involve multilingual interactions. Research highlights multilingual ASR as beneficial for SER pipelines due to better preservation of vocal cues across languages.

3. Noise-Robust Recognition
Call center recordings often contain background noise, telephone compression, and variable microphone quality. Whisper's strong noise resilience demonstrated in recent evaluations of Whisper-based emotion systems supports more reliable feature extraction for MFCC-based SER studies.

4. Emotion Detection
The system includes machine learning–based emotion classification aligned with established SER literature, which identifies emotion recognition as effective for analyzing stress, frustration, and other emotional states from vocal cues.

5. MFCC + Machine Learning Compatibility
The architecture is designed to work seamlessly with MFCC features,

acknowledged as one of the most stable and accurate feature extraction methods in SER research, and consistently used in call center–specific emotion studies.

6. Low-Latency Processing
Lightweight classifiers such as SVM, Random Forest, and KNN facilitate efficient processing, matching findings from real-time SER systems that highlight the importance of computational efficiency in emotion analysis applications.

7. Open-Source Accessibility
Whisper, SpeechBrain-related tools, and MFCC/ML frameworks used in SER research are open-source, enabling transparency, reproducibility, and customization. Open-source support is frequently referenced in SER literature as essential for experimentation and deployment.

8. Offline CSR Analysis Capability
Unlike cloud-dependent solutions, the proposed system operates completely offline, aligning with studies emphasizing the feasibility and importance of offline SER systems for privacy, data security, and real-world deployment in sensitive environments such as call centers.

Table 1 compares the functional capabilities of several speech processing and emotion recognition tools with the proposed Whisper-Based CSR Emotion Detection and Intervention System. While existing systems provide strong features in transcription, multilingual support, or acoustic feature analysis, none integrate all required components for CSR-centered emotional load assessment.

The proposed system uniquely combines high-accuracy Whisper ASR, MFCC-based emotional feature extraction, and lightweight machine learning (SVM, RF, KNN) to classify CSR affective states into Regulated Affective State, Elevated Affective Load, or High Affective Load. It additionally generates psychology-aligned supervisory recommendations, operates entirely offline, and is specifically designed for CSR not customer emotion monitoring. Existing tools such as Whisper and VOSK excel in transcription but lack emotion intelligence; SpeechBrain and openSMILE provide emotion features but lack integrated ASR; SenseVoice offers SER capabilities but is not optimized for CSR workforce assessment. Thus, the proposed system provides the most comprehensive and CSR-specific solution among all systems compared.

Table 2. Literature Matrix [LOCAL AND INTERNATIONAL]

Reference	Description	Strength	Weakness
[16] Montalbo (2022)	Examined burnout levels among Philippine call center agents; revealed high exhaustion due to customer hostility and workload.	Strong local relevance; highlights emotional strain among CSRs.	No technology or speech-based solutions.
[17] Charoensukmongkol (2021)	Studied mindfulness and emotional exhaustion among Filipino CSRs.	Introduces psychological frameworks relevant to CSR well-being.	No acoustic or AI analysis.
[18] Villacarlos (2022)	Identified predictors of psychological well-being among Philippine call center agents.	Provides insight into workplace stress factors.	Lacks objective emotional monitoring tools.
[9] Marshall (2024)	Investigated burnout and engagement in call centers; emotional strain affects performance and turnover.	Strong support for need of monitoring CSR emotions.	HR-centric; not technology-driven.
[1] Płaza et al. (2022)	Proposed emotion recognition for call center systems using acoustic features.	Validates SER using call recordings.	Customer emotion only; not agent-focused.
[2] Yurtay et al. (2024)	SER applied to call center audio using feature extraction methods.	Effective in phone-quality speech.	Does not analyze CSR emotional states.

[3] Harshitha et al. (2023)	Surveyed deep learning methods for SER.	Confirms MFCC & ML effectiveness.	Not CSR or workplace-specific.
[4] Dabbabi et al. (2024)	Introduced self-supervised SER techniques.	Explores advanced frameworks.	High computational demands.
[5] Alhussein et al. (2025)	Meta-analysis of conversational SER.	Demonstrates SER robustness across contexts.	No focus on workplace affect monitoring.
[6] Barhoumi et al. (2024)	Real-time SER with deep learning.	Shows feasibility of fast processing.	Requires heavy hardware.
[12] Guo et al. (2022)	MFCC + ensemble learning for SER.	Improved accuracy with MFCC.	No ASR integration.
[13] Alsabhan (2023)	SER via ensemble techniques.	Robust model performance.	Limited application to CSR workflows.
[14] George & Ilyas (2025)	Whisper embeddings used for SER.	Whisper improves noisy-audio emotion accuracy.	Does not map emotions to agent work states.
[15] Wewelwala & Sumanathilaka (2025)	Hybrid acoustic-textual SER for interactions.	Supports ASR+SER pipeline design.	Customer interaction focus only.
[19] Ekman (1999)	Identified six universal emotions.	Provides standardized emotional taxonomy.	Not speech-specific.

[20] Hekiert et al. (2022)	Analyzed acoustic stress markers such as pitch and vocal tension.	Validates prosody as emotional indicator.	Not designed for call center audio.
[21] Larrouy-Maestri et al. (2022)	Studied prosodic correlates of emotional expression.	Supports MFCC and prosody modeling.	Not workplace-specific.
[22] Opladen et al. (2023)	Linked prosody to emotional and cognitive load.	Reinforces emotional load classification.	Laboratory-based, not BPO-based.
[23] García-Madurga (2024)	Discussed risks and applications of Emotion AI in workplace settings.	Strong ethics and AI governance context.	Not audio-focused.
[24] Roemmich (2024)	Examined implications of AI emotion monitoring in work environments.	Highlights workplace practicality and risks.	No SER implementation.
[25] Mantello (2023)	Explored affective surveillance technologies.	Provides broader Emotion AI context.	Not technical in signal processing.
[26] Meng et al. (2024)	Reviewed affective computing tools for employee well-being.	Supports use of AI for workplace wellness.	Does not address speech signals.
[27] Kansal & Kumar (2024)	Acoustic stress detection using prosodic features.	Reinforces MFCC + prosody as stress markers.	No integration with ASR systems.

A.Local Studies

Local studies on the Philippine BPO industry consistently highlight the high levels of stress, emotional exhaustion, and burnout experienced by call center agents. Montalbo examined the burnout level of call center agents in Metro Manila and found elevated levels of exhaustion, cynicism, and reduced professional efficacy, particularly among younger agents, suggesting that emotional fatigue is a pervasive issue in the Philippine call center sector [16]. Charoensukmongkol investigated the relationship between mindfulness and emotional exhaustion among call center agents in the Philippines and reported that higher mindfulness was associated with reduced emotional exhaustion, emphasizing the psychological burden of call center work and the need for supportive interventions [17]. Similarly, a study on predictors of psychological well-being among call center agents in the Philippines identified job demands and psychosocial stressors as major contributors to burnout, emotional fatigue, and reduced well-being [18].

In addition to Philippine-based studies, Marshall analyzed burnout and employee engagement among customer service representatives in call centers and found that chronic emotional strain, customer hostility, and performance pressure significantly undermine engagement and job satisfaction [9]. Collectively, these local and context-relevant studies demonstrate that CSRs are at high risk of emotional overload and burnout. However, no local literature proposes an AI-driven, voice-based system for monitoring CSR emotional states. Existing research focuses largely on psychological, HR, and organizational interventions, rather than on automated speech-based emotion analysis using acoustic

features and machine learning. This gap provides a strong justification for developing a CSR-centered emotion detection and recommendation system based on call recordings.

B.Foreign Studies

Foreign literature offers a broad technical foundation for Speech Emotion Recognition (SER) and the analysis of emotions from speech and prosody. Płaza et al. and Yurtay et al. developed emotion recognition methods for call center systems using acoustic features extracted from call recordings, demonstrating that emotional states can be inferred from telephone-quality speech, though their systems primarily focused on customer emotions rather than CSR states [1], [2]. Comprehensive surveys by Harshitha et al., Dabbabi et al., and Alhussein et al. reviewed SER techniques and concluded that Mel-Frequency Cepstral Coefficients (MFCC) remain among the most effective features for detection, and that lightweight classifiers such as SVM, Random Forest, and KNN continue to perform strongly in many SER tasks [3]–[5]. Real-time SER implementations by Barhoumi et al. and Alsabhan further showed that emotional analysis can be integrated into interactive systems using deep learning and ensemble techniques [6], [13].

Guo et al. demonstrated that combining MFCC with ensemble learning can significantly improve SER accuracy [12], while George and Ilyas explored Whisper-based embeddings for speech emotion recognition and reported enhanced robustness in noisy environments [14]. Wewelwala and Sumanathilaka proposed hybrid acoustic–textual models for improving customer interaction analysis, indicating the growing trend toward combining ASR outputs and acoustic features for richer emotional modeling [15]. Research

on vocal indicators of stress and prosody further supports this work: Ekman's foundational theory on basic emotions formalized six universal emotions happiness, sadness, anger, fear, disgust, and surprise which continue to guide emotion classification frameworks [19]. Studies on emotional prosody and vocal stress markers, such as those by Hekiert, Larrouy-Maestri, Opladen, and others, show that pitch, intensity, speech rate, and prosodic variability are reliable acoustic correlates of emotional arousal and stress, reinforcing the validity of using MFCC and related features for vocal emotion assessment [20]–[22].

In parallel, foreign literature on AI and workplace well-being emphasizes both the potential and the risks of emotion-aware systems. Valtonen et al. examined AI and employee well-being and found that AI tools can indirectly influence well-being through changes in work processes and task structures [10]. Additional studies by García-Madurga, Roemmich, Mantello, Meng, and others discuss Emotion AI and affect recognition in workplace environments, highlighting opportunities for monitoring well-being as well as ethical concerns regarding surveillance, privacy, and emotional misclassification [23]–[27]. Together, these works establish a strong foundation for using acoustic features, machine learning, and ASR in emotion recognition and workplace applications. However, they also reveal a clear research gap: there is no existing system specifically designed to analyze CSR emotional load from call recordings, classify it into psychology-aligned occupational states (such as Regulated Affective State, Elevated Affective Load, and High Affective Load), and generate non-diagnostic supervisory recommendations. The proposed system aims to fill this gap by combining Whisper ASR, MFCC-based SER, and lightweight machine learning within a CSR-centered, ethically aware framework.

B. Conceptual Framework

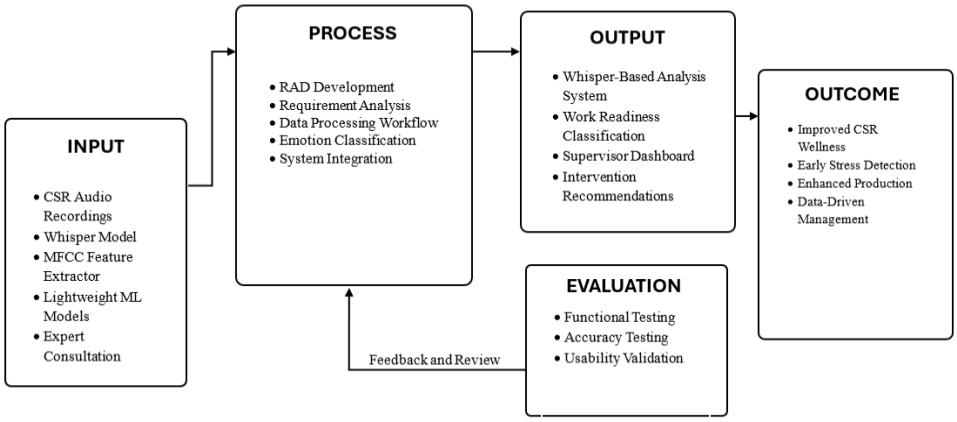

Figure 5 Conceptual Framework

The conceptual framework of the Intelligent Whisper-Based Call Recording Analysis System for CSR Emotion Detection and Automated Intervention Recommendations Using MFCC and Lightweight Machine Learning illustrates how the components of the study interact from initial requirements to the final evaluation of the developed system. It follows the Input–Process–Output–Outcome–Evaluation (IPOOE) structure required in the program.

The Input stage contains all preliminary requirements needed to develop the system. These include CSR audio recordings, the Whisper ASR model for transcription, the MFCC feature extraction method for capturing emotional acoustic characteristics, lightweight machine learning classifiers (SVM, RF, KNN), development tools, and expert consultation. These elements serve as the foundational resources for designing and implementing the system's emotional analysis pipeline.

The Process stage represents the research methodology used in developing the system, which follows the Rapid Application Development (RAD) model. This includes requirements analysis, iterative prototype refinement, and the complete data-processing workflow composed of audio transcription, MFCC feature extraction, and emotion classification. It also includes system integration, where the backend processing, machine learning components, database, and dashboard interface are combined into one operational platform. These processes transform raw audio into structured emotional indicators.

The Output of the framework is the completed information system: a fully functional Whisper-based analysis tool capable of transcribing CSR speech, extracting emotional features, classifying emotional load into occupational categories, and generating automated intervention recommendations. The output also includes a supervisor dashboard that displays transcripts, emotional results, historical trends, and recommended actions.

The Outcome reflects the expected benefits resulting from the use of the system. These include improved CSR emotional wellness through early detection of stress, enhanced managerial decision-making based on objective emotional cues, increased workplace productivity, and the promotion of data-driven operational management within BPO settings. These outcomes align directly with the problems identified in the literature regarding CSR emotional fatigue and the lack of automated monitoring tools.

Finally, the Evaluation stage ensures the reliability and effectiveness of the system through functional testing, accuracy testing of the emotion classification models, usability

validation with end-users, and repeated refinement based on evaluation feedback. The feedback loop in the framework indicates continuous improvement, as insights from testing are used to update and enhance the system before full deployment.

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C. Definition of Terms

Term1 Operational definition of Term 1, how was Term1 used in the study. Operational definition of Term 1, how was Term1 used in the study.

Term2 Operational definition of Term 2, how was Term2 used in the study. Operational definition of Term 2, how was Term2 used in the study.

Chapter 3

TECHNICAL BACKGROUND

A. Software Development Requirements

Table 1 Software Requirements

SOFTWARE	DESCRIPTION

Table 3 contains all the software that you will be using including their names and description. Explain here how you will use the software for your proposal. Table numbering should be continuous from Chapter 1 to Chapter 3. Adjust accordingly.

B. Hardware Development Requirements

Table 2 Hardware Requirement

HARDWARE	SPECIFICATION

Table 4 contains all the hardware that you will be using including their names and specifications. Explain here how you will use the hardware and their purpose in your proposal.

C. Network Architecture

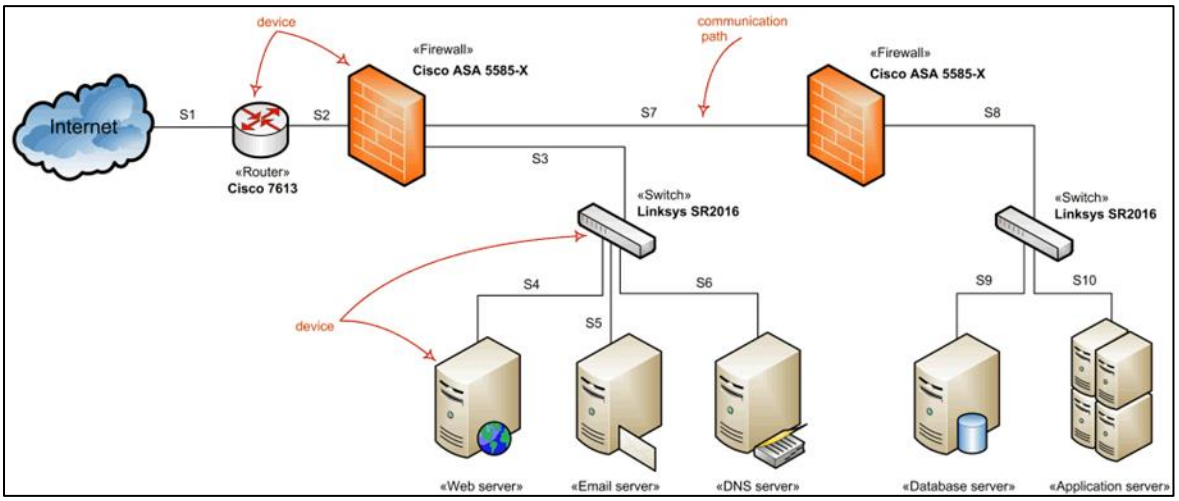

Figure 1 Network Architecture

Figure 2 shows the proposed network architecture. Describe here how the network components are interconnected. This section is required for NA students but on a case to case basis for other specialization. This will depend on the proposal.

D. Sources of Data for Development and Testing

Table 3 Sources of Data

Sources	Description
Source	Type of data, year of acquisition, and other pertinent details.
Source	Type of data, year of acquisition, and other pertinent details.

CHAPTER 4

METHODOLOGY

A. Research Design

Since you will be developing software along with your chosen algorithm. Specify in this section what Software Development Life Cycle (SDLC) you will follow and discuss the steps involved in the chosen SDLC.

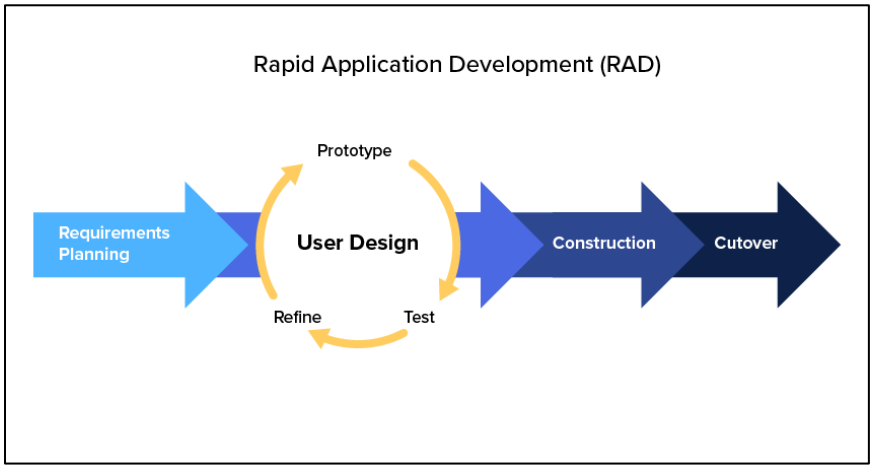

Figure 2 Sample SDLC(RAD)

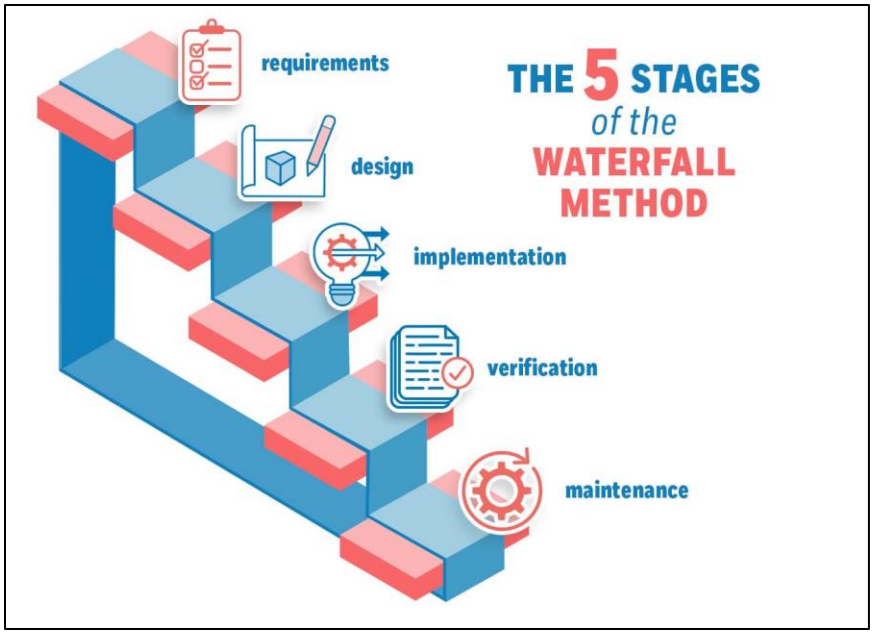

Figure 3 Sample SDLC (Waterfall)

B. Diagrams

Compile and discuss diagrams pertaining to the development of the system being proposed. See to it that all are properly labeled and discussed.

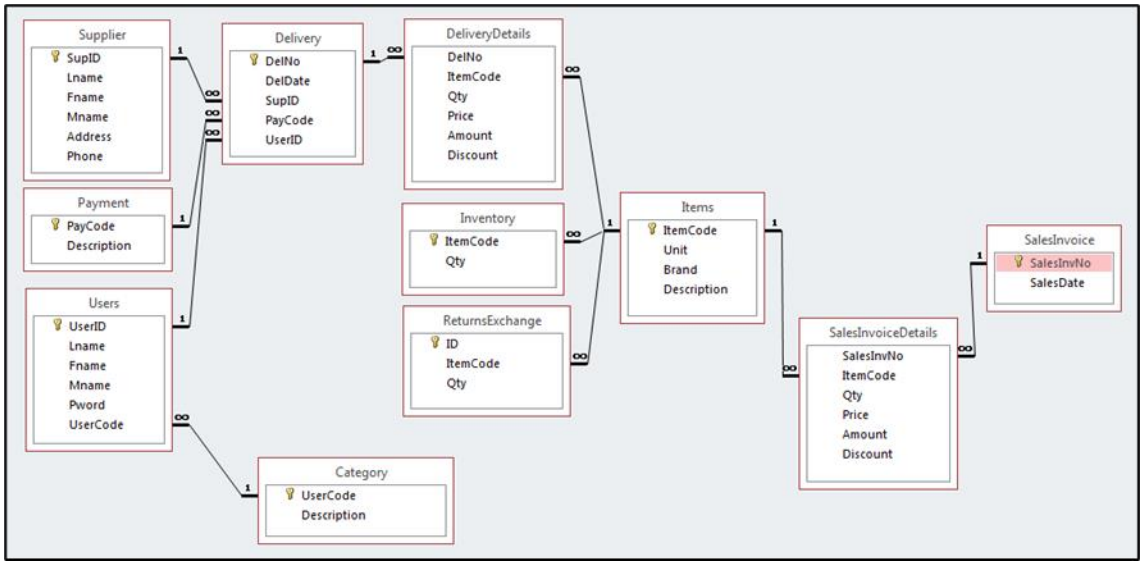

Figure 4 Database Schema

Figure 5 displays the intended Database Schema. Discuss considerations and reasoning behind the diagram

C. Block Diagram of <system> using the <?> algorithm

Here you will draw the block diagram of how the algorithm(s) will be used in the proposed system. This area is vital to show you fully understand what you will be applying in the development of the system. <See sample below>

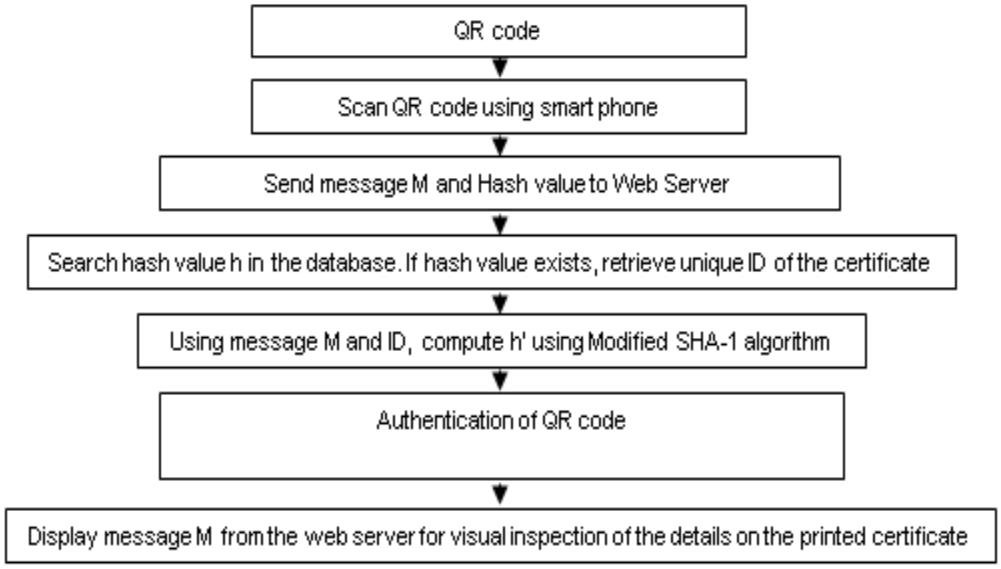

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Figure 5 Block Diagram of <system>using the < >algorithm

Figure 8 shows the block diagram of this process. Users who want to verify the certificate will use their smartphones running on Android with a camera to scan the QR code. After

capturing the QR code, the web application computes the hash value and send the hash and the message to the web server. After receiving the information, the server search for the hash. If the value exists, the system will retrieve the unique ID of the certificate from the database. Using message M and ID, the modifiedSHA-1 compute for the hash value h'. If hash value h' = hash value h, the message is said to be authentic. Then, the system will display the resulting message for visual inspection and compare this to the details of the printed certificate. Otherwise, we can say that the QR code message M is modified. *<Note: This paragraph is a sample only>*

D. Potential Ethical Considerations

Discuss here potential ethical norms include issues such as requirements for honesty, requirements for informed consent, anonymization and storage of data, the right of access to data for participants and duty of confidentiality for all those who undertake research.

E. Story Board

Here you will draw and discuss potential design considerations and constraints that you have. You can put design mock ups here to show how you visualize your proposed system.

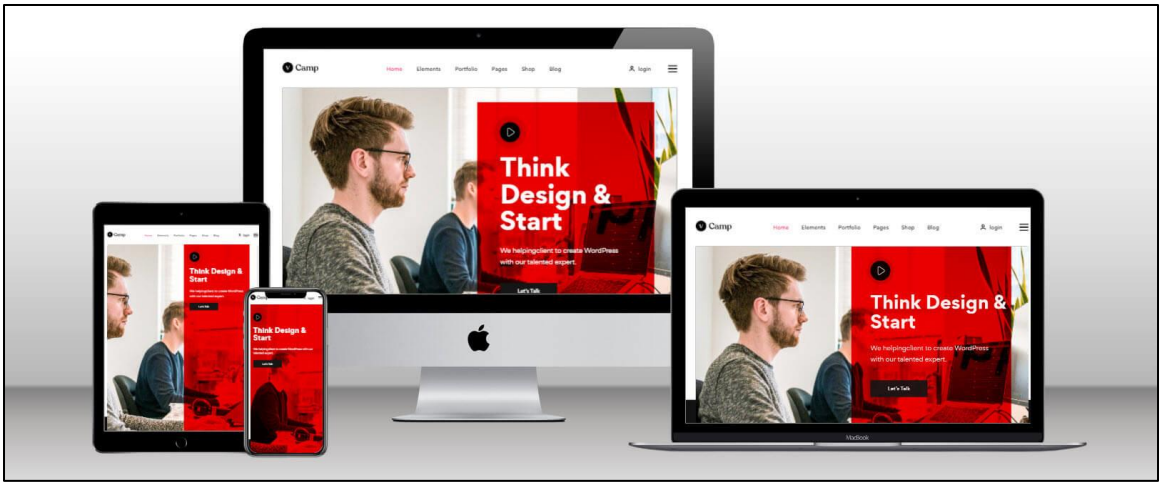

Figure 6 Design Mockup

References

[1] J. Chan, "This is a sample reference", 2023.

Appendix A (SAMPLE)